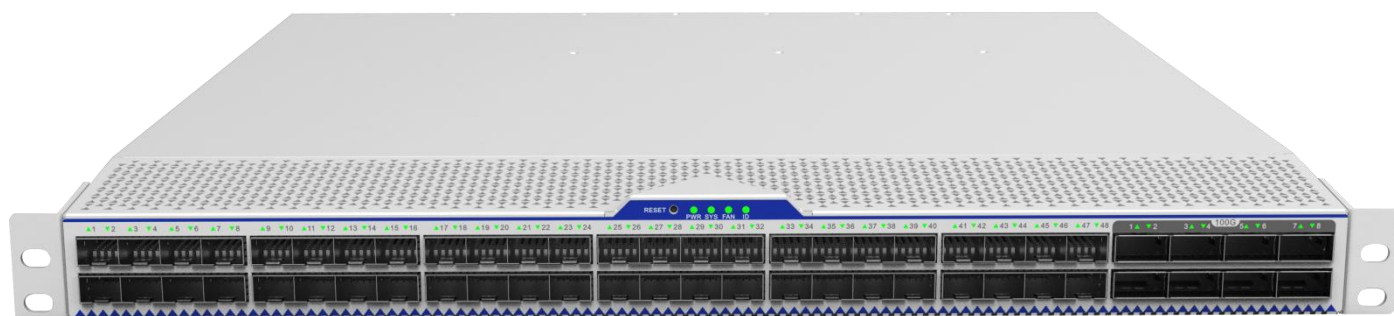


## Data Center Switch S6750-48HX8CQM-AC



S6750-48HX8CQM-AC TOR switch – with superior port density of 48×25 Gbps and 8×100 Gbps ports is ideally suited for deployment at the server access layer in large enterprise data centers.

S6750-48HX8CQM-AC delivers high-performance, high interface density and low latency to allow enterprises and carriers to build cloud-oriented data center networks. With port speeds spanning from 10Gbps to 100Gbps per port and port density that enables full rack connectivity to any server at any speed S6750-48HX8CQM-AC switch is an ideal leaf and top of rack (ToR) solution, allowing maximum flexibility.

### Hardware Specification

|                    |                                                                         |
|--------------------|-------------------------------------------------------------------------|
| Model              | S6750-48HX8CQ-AC                                                        |
| Management ports   | 1x RJ45 100/1000Base-T out-of-band<br>1x RJ-45 Console<br>1x USB Type A |
| Switch ASIC        | Marvell Falcon 98CX8514                                                 |
| CPU                | Intel Broadwell-DE D1508                                                |
| Memory             | Default 8GB DDR4<br>Max 32GB DDR4                                       |
| Storage            | 16GB eMMC NAND FLASH<br>SSD module optional                             |
| BMC                | AST2520 optional                                                        |
| Power supply units | 2 x 100-240 VAC, 550W<br>Redundant, hot swappable                       |
| Power consumption  | 520W max                                                                |
| Fan unit           | 5 (4+1 redundant)                                                       |

|                              |                                            |
|------------------------------|--------------------------------------------|
|                              | Hot swappable                              |
| Form factor and dimension    | 1 RU<br>(W) 445 mm x (D) 470mm x (H) 44 mm |
| Weight(NW)                   | 9.6 kg                                     |
| Operating temperature        | 0°C– 45°C                                  |
| Storage temperature          | -40°C– 70°C                                |
| Operating humidity           | 5% - 95% (non-condensing)                  |
| Storage humidity             | 5% - 95% (non-condensing)                  |
| Operating altitude           | 3000m max                                  |
| Safety regulation compliance | UL/CSA, cUL, CB                            |

### Performance and Scalability

|                                                                           |                                                                                                          |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Model                                                                     | S6750-48HX8CQ-AC                                                                                         |
| Switching Capacity                                                        | 4.0 Tbps                                                                                                 |
| Forwarding Rate                                                           | 2.9 Gpps                                                                                                 |
| Jumbo frames                                                              | Yes                                                                                                      |
| Packet Buffer Size                                                        | 24 MB                                                                                                    |
| Transmit Descriptors                                                      | 192K                                                                                                     |
| Bridge FDB Entries                                                        | Up to 256K                                                                                               |
| Router IPv4 Host Entries                                                  | Up to 256K                                                                                               |
| Router IPv6 Host Entries                                                  | Up to 128K                                                                                               |
| Router LPM                                                                | IPv4 - up to 512K unicast prefixes<br>IPv6 - up to 256K unicast 64-bit prefixes                          |
| TCAM Resources (Shared between TTI, IPCL and EPCL Classification Engines) | 36K x 10B rules (18K x 10B per dual control pipe)                                                        |
| Virtual Ports (ePorts)                                                    | 8K                                                                                                       |
| Bridge Domains (eVLANs)                                                   | Up to 8K                                                                                                 |
| Multicast Physical Port Groups (VIDX)                                     | Up to 4K                                                                                                 |
| Multicast Virtual Port Groups (eVIDX)                                     | 16K                                                                                                      |
| Multicast Linked List (MLL) Entries                                       | 16K                                                                                                      |
| Tunnel-Start/ARP Table                                                    | Up to 192K ARP entries<br>Up to 48K Non-IPv6 Tunnel-Start entries<br>Up to 24K IPv6 Tunnel-Start entries |

|                                                                  |                                                                    |
|------------------------------------------------------------------|--------------------------------------------------------------------|
| Router Next Hop Entries                                          | 24K                                                                |
| Router ECMP Groups                                               | 12K                                                                |
| ePort ECMP Groups                                                | 16K                                                                |
| Trunk ECMP Group                                                 | Up to 4K                                                           |
| L2 ECMP Table (Shared by ePort ECMP group and Trunk ECMP groups) | 16K ECMP members Each ECMP group may contain up to 4K EMCP members |
| Hardware OAM Flows                                               | 16K ECMP members 4K OAM flows (1K OAM flows per control pipe)      |
| Spanning Tree Groups                                             | Up to 4K                                                           |